

# Multimedia Classics
# Hypermedia Hermeneutics

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NODEM
Run Run Shaw
School of Creative Media
City University Hong Kong
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你们好

叨扰

I would like to thank the whole organizing team, first of all Sarah and Waltraut for inviting me to this wonderful conference in this amazing town.

And I would like to thank Swissnex for spending part of my travel costs.

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## Gesamtkunstwerk

## Neolithic age of Multimedia

## Silver disc classics

For more than 20 years online and offline applications of hypermedia technology have combined text, image, video, animation and sound into a total work of art, a "Gesamtkunstwerk".

Hypermedia applications comprehend online and offline productions like LaserDisc, Photo-CD, CD-i, CD-ROM, DVD-ROM as well as kiosksystems or mobile technologies like handheld guides.

Often produced for museums, the digital masterworks of applied art are icons, signs, and symbols of and for our digital and digitised cultural heritage.

And there are important relics of the "Neolithic age"of multimedia.

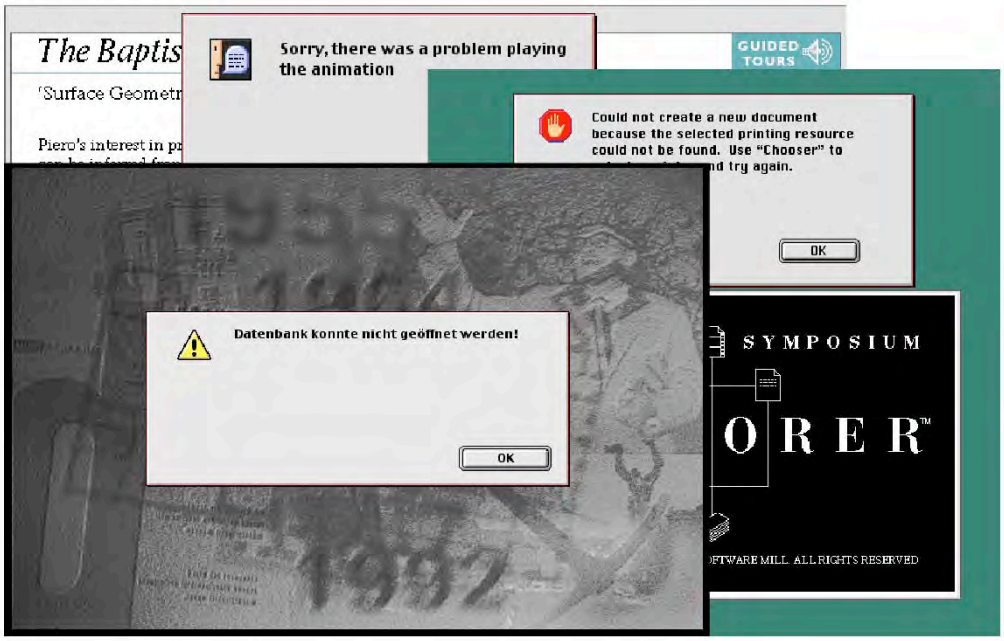
But its a pity that later generations will have no access on these marvellous "silver disc classics" as well as early kiosk systems and early museum webpages.

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National Gallery London: Art Gallery, 1993

Josef Albers: Interaction of Color, 1994

Documenta 1-9, 1997

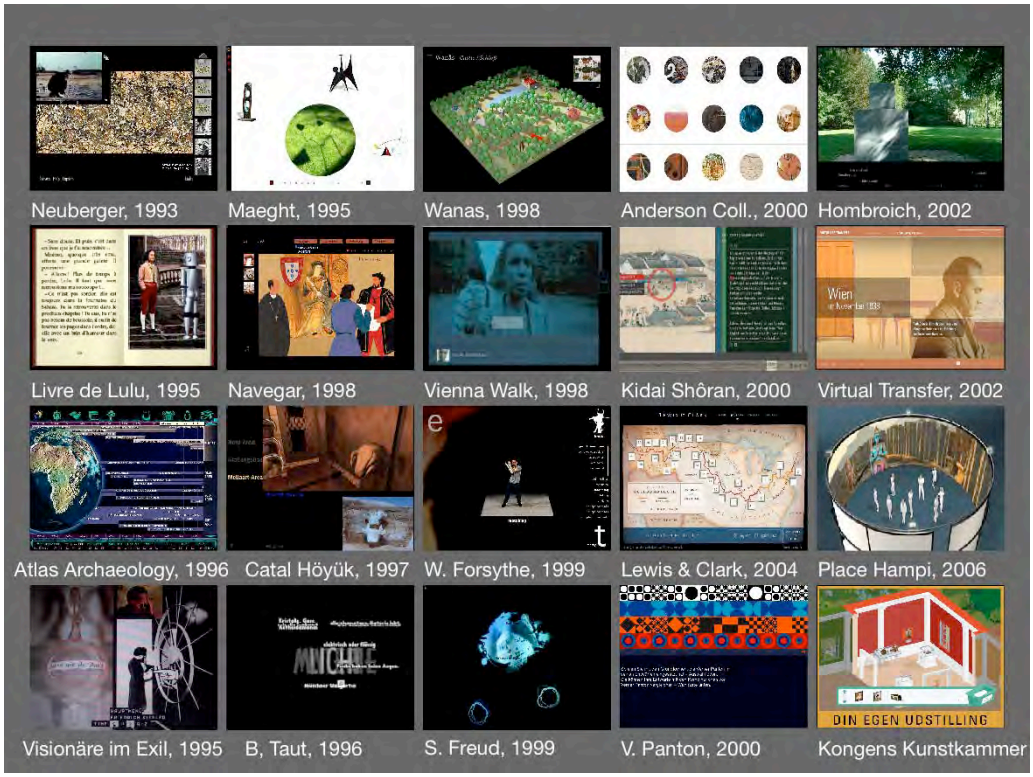



At the moment most of the CD-ROMs are still readable, but it is only a matter of time before one of the next computer generations disallows the access to the hypermedia classics. To use old CD-ROMs on Apple Macintosh Computers, the "Classic" environment (Mac OS 9) has to be installed.

Since the Apple Mac computers are running on Intel processors (Mac OS X and later), the installation of Classic is technically infeasible in defiance of diverse experiments with emulator programs. This is the reason why you see me here standing with this old-fashioned Mac PowerBook G4 Mac OSX Version 10.4.11.

Most of the archives or museum libraries that have collected hypermedia formats like CD-ROM, CD-i or Laserdisc no longer have computer systems running with the old specifications. Therefore they have stopped collecting these formats. Meanwhile, the technical specifications of these relics are powerful signs of the powerlessness of the next computer generation who will disallow access to the hypermedia classics.

But the soaring loss on accessibility is only one problem which we have to solve.




Hypermedia applications are hybrids. They are masterpieces of sampling information and cultural content in different forms by combining different strategies of dramaturgy, navigation and design. Therefore the similar objective targets can create completely different solutions. The results can look

like an interactive time table as on Atlas of Archaeology

like a comic novel as on Navegar, describing the discovery of the world by the Portugese voyagers,

like dancing words as the whispering voices of the ghosts on Bruno Taut

like a walk in the park as on Museuminsel Hombroich or

like a jogging avatar in the Castle Woods of Wanås.

Previous efforts to construct a typology for such hypermedia applications are often a mixture of content and application area. They condemned to failure because they have disregarded this variety of possible applications. So a simple classification cannot follow all terms and conditions.



What can we learn from the *Pilgrim Fathers* of Multimedia for the future?

Why it is so difficult to describe and analyse the interplay of content, navigation, and design of hypermedia applications?

How can the complexity of knowledge and of our cultural heritage be transferred and structured by the instruments of hypermedia?

And what exactly is the polyvalence (the different levels of value) for knowledge transfer by using hypermedia?

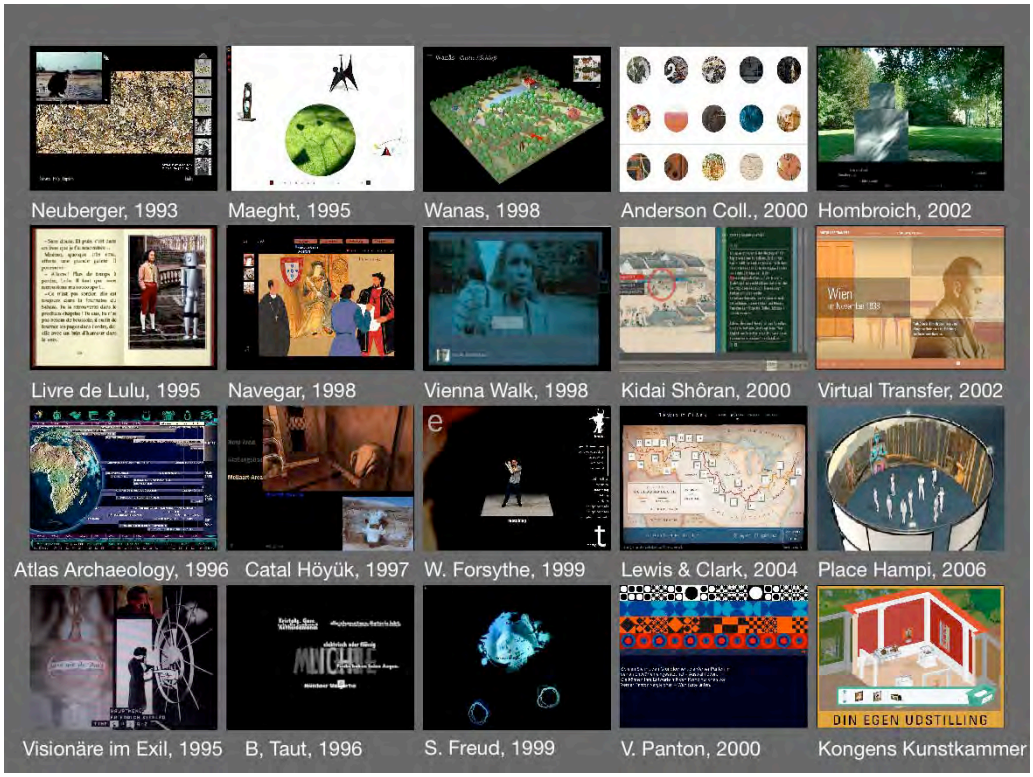
There are a lot of questions. So, let's concentrate in a few:

What can we learn from the *Pilgrim Fathers* of Multimedia for the future?

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Hypermedia will change the perceptions of the museum visitor and the strategies of museum communications. Hypermedia is open, transient, interdisciplinary, process-related, discursive, and dependent on concept and context. Hypermedia is digital, multimedia-based and extended by the interactivity with its recipient, the user. Hypermedia is so diverse that it is in need of documentation, analysis, and methodology in a wider sense. In this respect hypermedia applications are comparable with works of contemporary art. The challenge is to reinterpret our traditional understanding of information as well as our understanding of knowledge transfer. At the moment information about hypermedia in museums is mostly published as project-oriented articles in conference proceedings. Meanwhile there are countless conferences and workshops hold every year producing tons of grey literature. But there is lacking an analytical study in the sense of a scientific methodology as well as an oeuvre catalog with exemplary case studies for online and offline hypermedia applications in museums until today. What is missing is a critical canon.

Dramaturgy in Museum Communication: The Change of Meaning in Virtual Spaces
Swiss National Museum, Zurich 2003

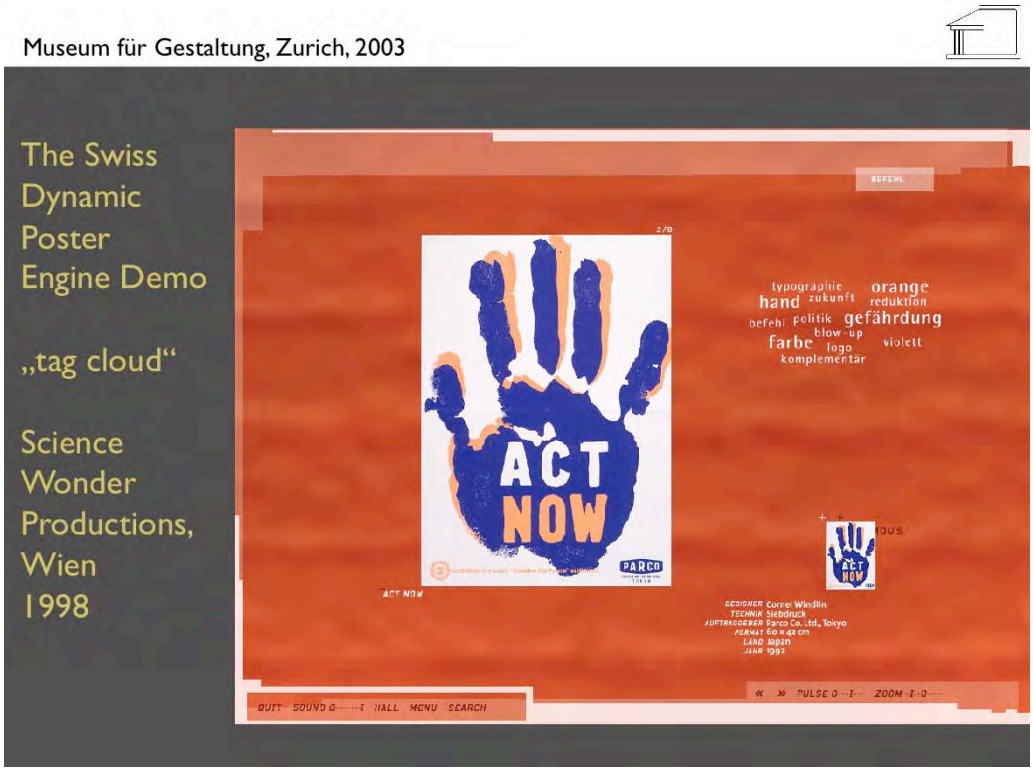


It looks that hypermedia will be a perfect terrain for research and therefore I started my project after discussing it with some colleagues like Xavier Perrot, Peter Samis, Holly Witchey, and Sarah Kenderdine on the symposium "Dramaturgy in Museum Communication: The Change of Meaning in Virtual Spaces", which I have organised during my time at the Swiss National Museum in Zurich.




Now is the right moment to tell you a little bit about myself and my motivation behind this project.

In the last few years I've produced a lot of different hypermedia applications. Working as an art historian in that field means to me to understand and analyse the process of creating new knowledge with the possibilities of digital technologies.



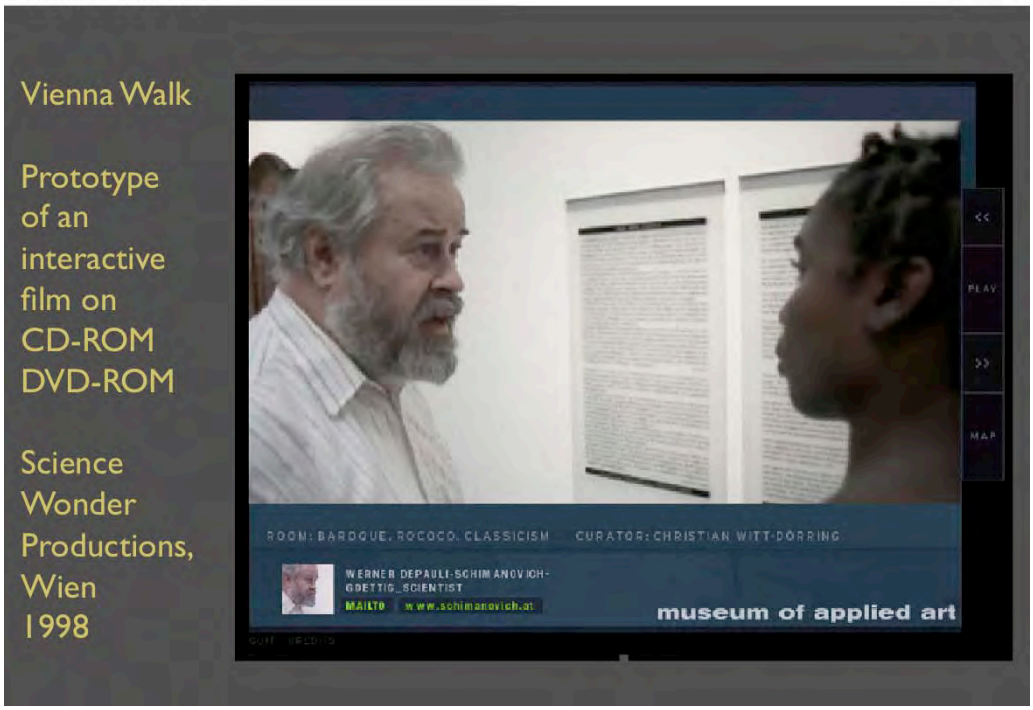

Here is a screen shot of the database The Swiss Dynamic Poster Engine Demo on CD-ROM. As you can see we have used tag clouds in 1998 to create keywords for the poster collection of the Design Museum in Zurich.

Vienna Walk, 1998




In the same year we have produced Vienna Walk Demo. This prototype of an interactive film on DVD-ROM combines the possibilities of digital film, hypermedia and the web. In 1998 it was a futuristic vision of real interactivity and an intelligent knowledge transfer in the field of cultural heritage-tourismecommerce.

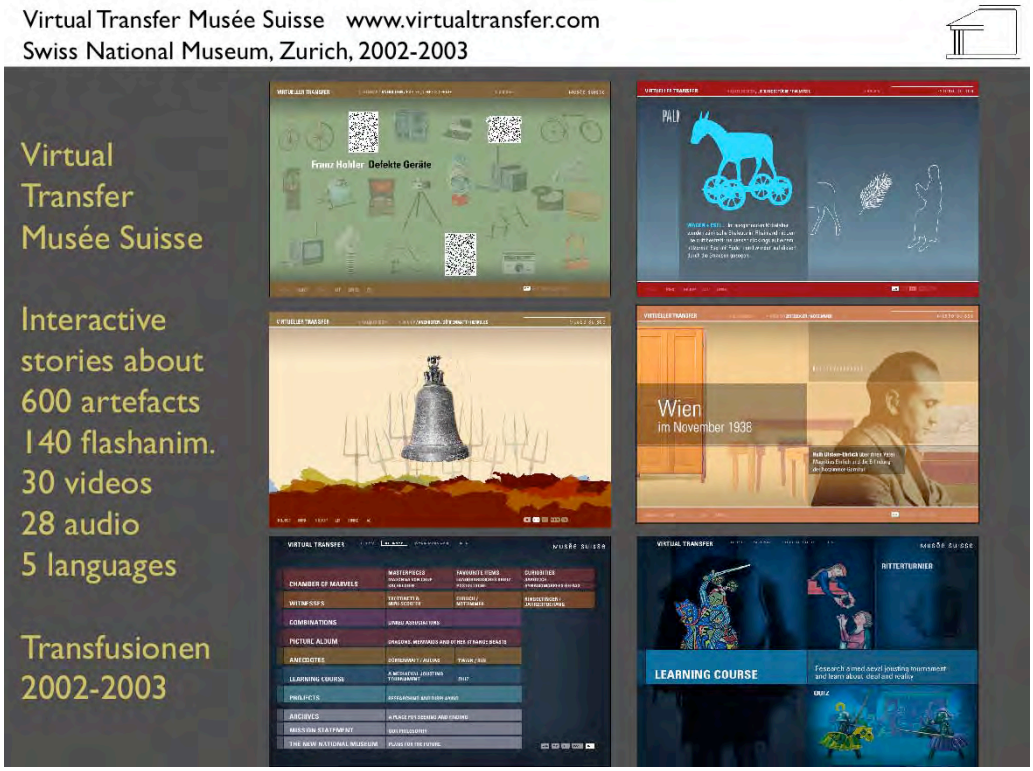
Vienna Walk, 1998




Here you can see agent Pauline as a living avatar asking the natural scientist Werner Schimanovich what is the purpose of art. If you don't like his answer, you can send him an email without leaving the film. This is real interactivity. For Peter Samis Vienna Walk is the beginning of Social Media.




For the Museum of applied Arts in Vienna I've realised the first CD-ROM about Historicism and the pluralism of styles in the 19. century.



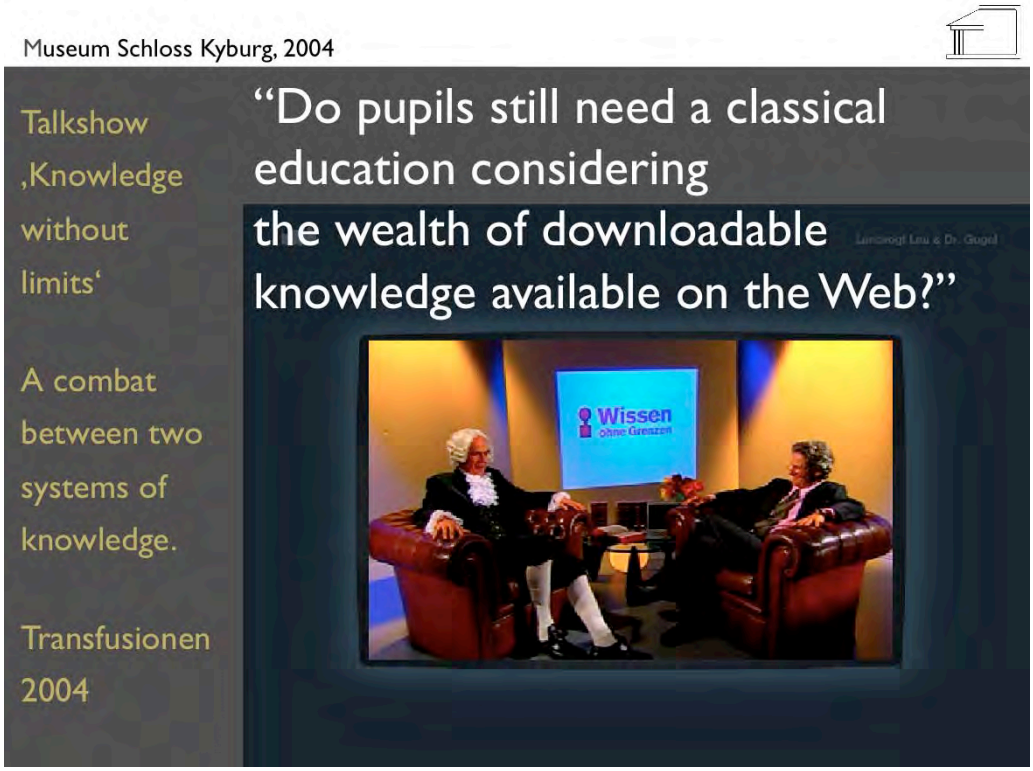

The VIRTUAL TRANSFER MUSEE SUISSE is a strategy for online communication by interactive narratives in five languages (German, French, Italian, Romansh and English) for the Musée Suisse Group. The objects of the museum collections became the narrators and the stories they have to tell are part of the visitor's own life, like tolerance, xenophobia or transitoriness. Different strategies and dramaturgies have been used to explain the deeper meaning of the artefacts.




On the CD-ROM Museum Schloss Kyburg no other than the fictional moderator Dr. Gugel has invited the famous Swiss encyclopedist Johann Jacob Leu to a fictional TV talk show named "Knowledge without limits".

Leu and Gugel are discussing the different methods of attaining knowledge in their respectable centuries.

The discussion takes up the following controversial question:

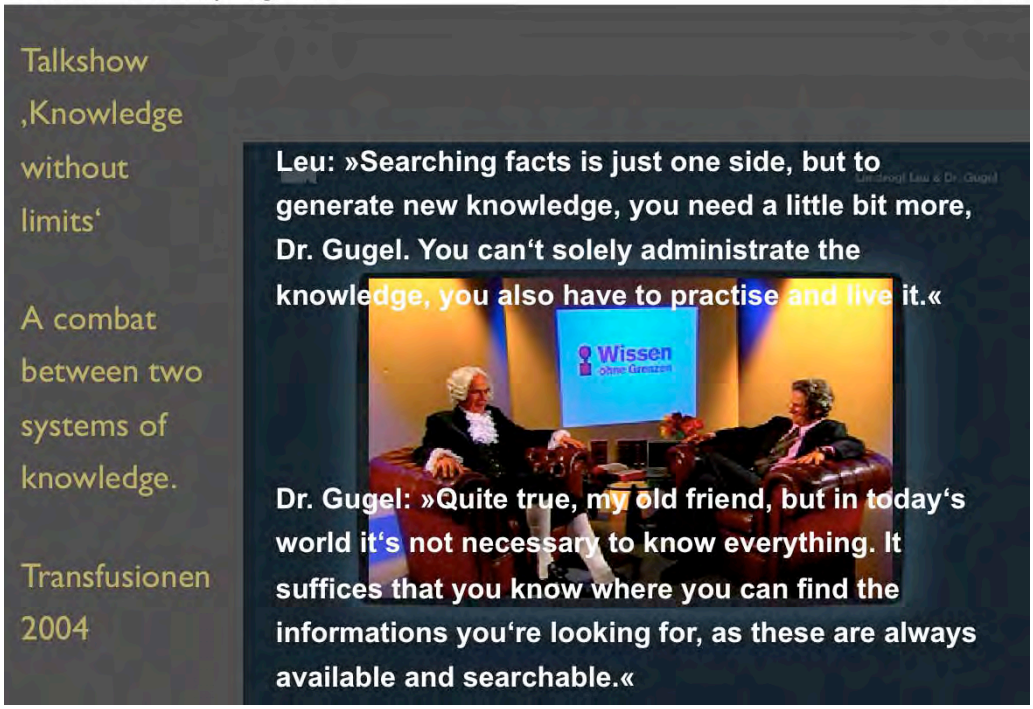



"Do pupils still need a classical eduaction considering the wealth of downloadable knowledge available on the Web?"

In a special competition Leu's knowledge-based lexicon competes against Dr. Gugel's search engine.

Then both discuss the pros and cons of the different methods:

Museum Schloss Kyburg, 2004




Leu: »Searching facts is just one side, but to generate new knowledge, you need a little bit more, Dr. Gugel. You can‘t solely administrate the knowledge, you also have to practise and live it.«

Dr. Gugel: »Quite true, my old friend, but in today‘s world it‘s not necessary to know everything. It suffices that you know where you can find the informations you‘re looking for, as these are always available and searchable.«

By following Leu and Dr. Gugel, the user has learned that both systems of knowledge-based methods have theirr positive and negative aspects.



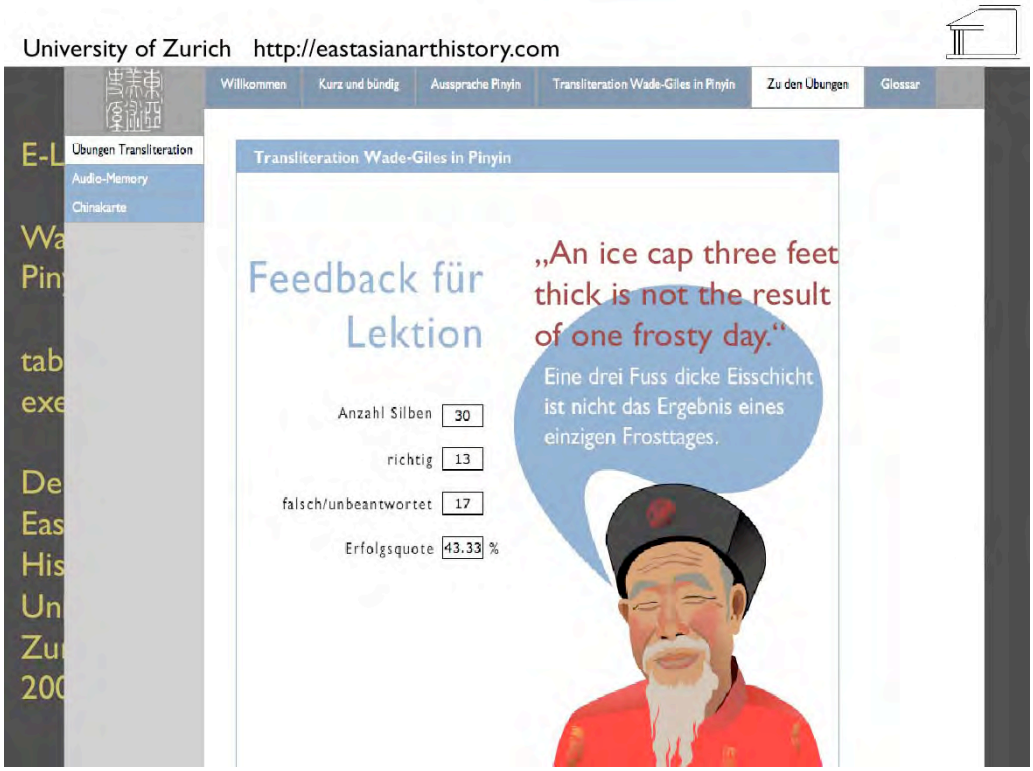

For the Department of East Asian Art History of the University Zurich I've produced a learning course about Hanju Pinjin.

This means how to speak and write Pinjin and how to transliterate Wade-Giles into Pinjin.

Master Kungci, the famous Confucius himself will be your teacher and believe me, he looks mild, but he is a very strong teacher.

You will find table with all character, how to write and how to speak it. And there a lot of exercices with different levels for the rookie as well as the champion.

After surviving the test Master Kungci will give you advice:

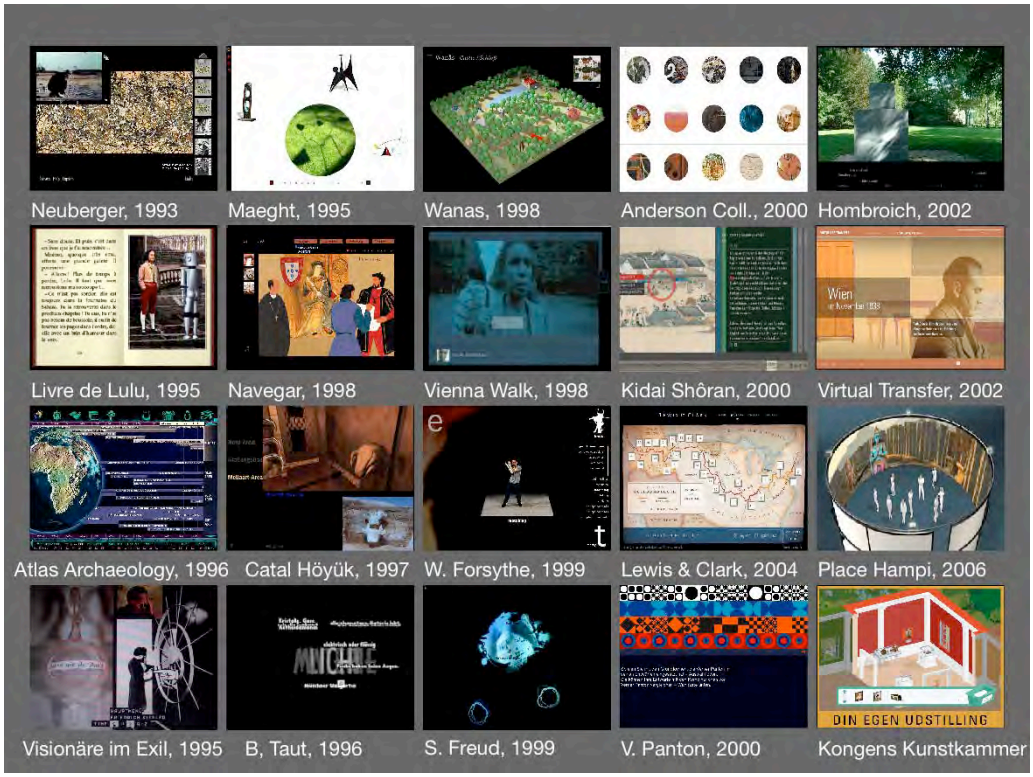



„An ice cap three feet thick is not the result of one frosty day.“

Which means: Try again and this time try harder.




In another part of the course the students will find a map to learn the names of the provinces in China and 40 different audio memories to test their ability to combine the character with the pronounciation.




As you can see I have a lot of fun producing these things. So I combine the strength of analytical thinking with the joyful designing of hypermedia applications together with a certain scientific curiosity, my passion to teach, and the ability of a critical eye.

Since 1993 I'm collecing online and offline hypermedia applications and in the last few years I have analysed over 700 of them, entererd their data into a dbms, developed a methodology for hypermedia documentation and I am still working on the creation of an oeuvre / a work catalog of case studies with 101 Multimedia classics.

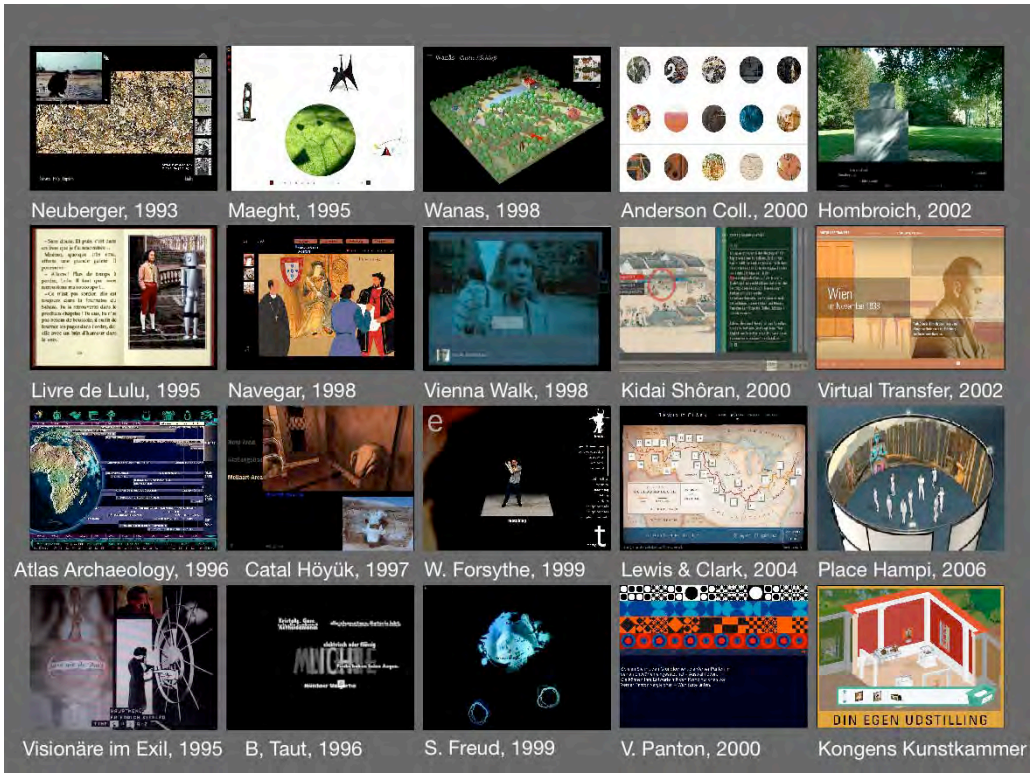
Part of the database will be accessible on the web. But the result of this work will be first of all a book. It may be strange to write a book about Hypermedia applications, but „it must be a book, because it is an instrument at the time of appearance and later it will become a document". (Lars Müller). And for a lot of our colleagues in the academic world it must be a book to make the invisible visible.



„Objects are for perception,
works of art for emotion
and media for fun.
Media are - first of all - for fun.“

Krzysztof Pomian, June 4th 2009

"Objects are for perception, works of art for emotion and media for fun. Media are – first of all – for fun."
With this statement, the famous museologist Krzysztof Pomian had given a statement about media, which is for a lot of museum people more than true. Not every curator is happy about the usage of media in his showroom.




In the last years scientific research in the visual studies has been concentrated mostly on media art. Only if a commented oeuvre catalogue as a basic work will exist, hypermedia applications in museums will become as a kind of "hypermedia archaeology" a new recent field of research in the visual media sciences and art media history. Digital Media archaeology will be one of the sciences of our future.

The whole project will give impulses to museology, museum communication as well as art media history, media sciences and other related disciplines. Hypermedia is influencing more and more the scientific research, the academic teaching as well as the communication in museums and between museums, archives and libraries.

By using hypermedia applications in the teachings the students will learn how to create a storyboard, navigate into knowledge and design cultural heritage. The gap between curators, scientists, media designer and programmer can be bridged, new application areas like content provider in a museum or information designer of cultural heritage will be supported.

**Harald Kraemer MULTIMEDIA CLASSICS - HYPERMEDIA HERMENEUTICS**

1. Introduction
Hypermedia - fine or applied art? / State of research / New Digital Heritage
2. Hypertext, Multimedia, and Hypermedia (definition)
3 ŒUVRE CATALOGUE (101 Multimedia Classics)
4 FUNCTION & FORM (analysis)
Typologies / Multifunctionality / Systemic, Scenic, or Scenographical Design
5 CONTENT
THE IMPACT OF THE PROLOGUE
THE VALUE OF FACTS
THE ART OF NARRATIVE
EXPLORATIVE DISCOVERIES
THE LUDIC APPROACH
THE PURPOSE OF EDUCATION
6 POLYVALENCE (the many levels of value)
Audio-visual Rhetoric / Hypermedia Grammar
Gesamtkunstwerk / Pendantsystem / Look & Feel Good
7 INTERDEPENDENCE & CONSEQUENCE
Authenticity & Credibility / Reflection & Perspectives
8 Notes / References / Bibliography / Index

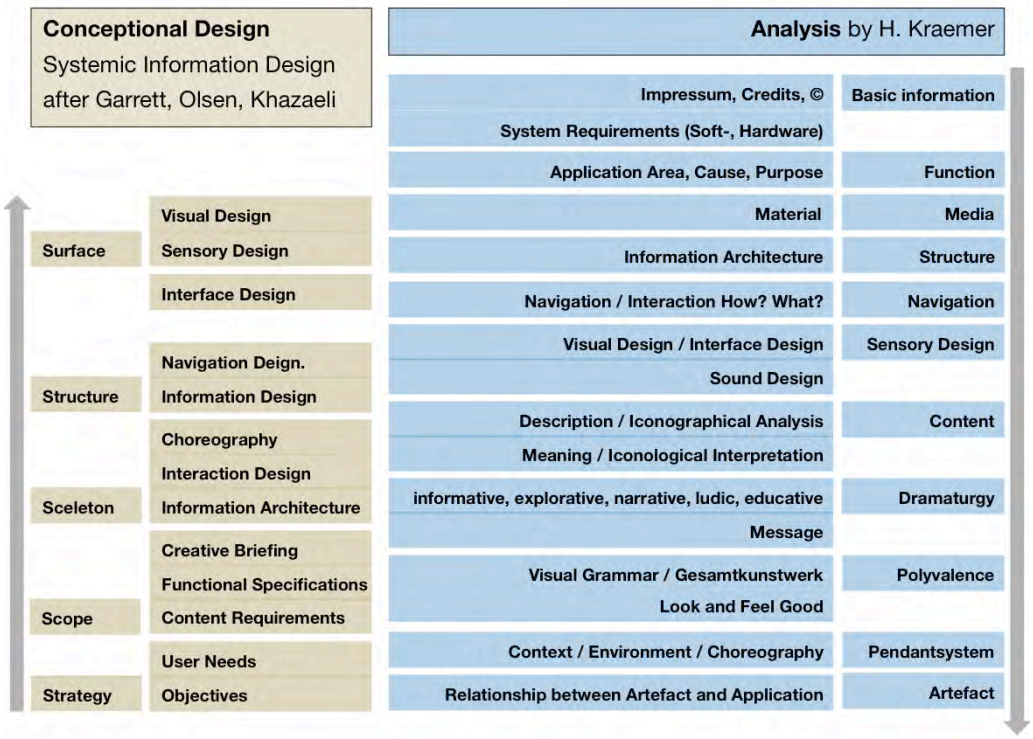
So, here are the Contents of the book.

As you can see, it will be not possible for me to talk about every chapter and subchapter. Therefore I will concentrate on two parts.

First, will be an overview into my method of analyzing Hypermedia.

Second, I would like to give an introduction into the chapter Polyvalence.

Unfortunately I can't give you an answer to the question: Is there a correlation between the behavior of a museum visitor, the user of hypermedia applications and the information architecture of the application itself? And, if yes, what may be the philosophical background behind it? There will be no time for this.

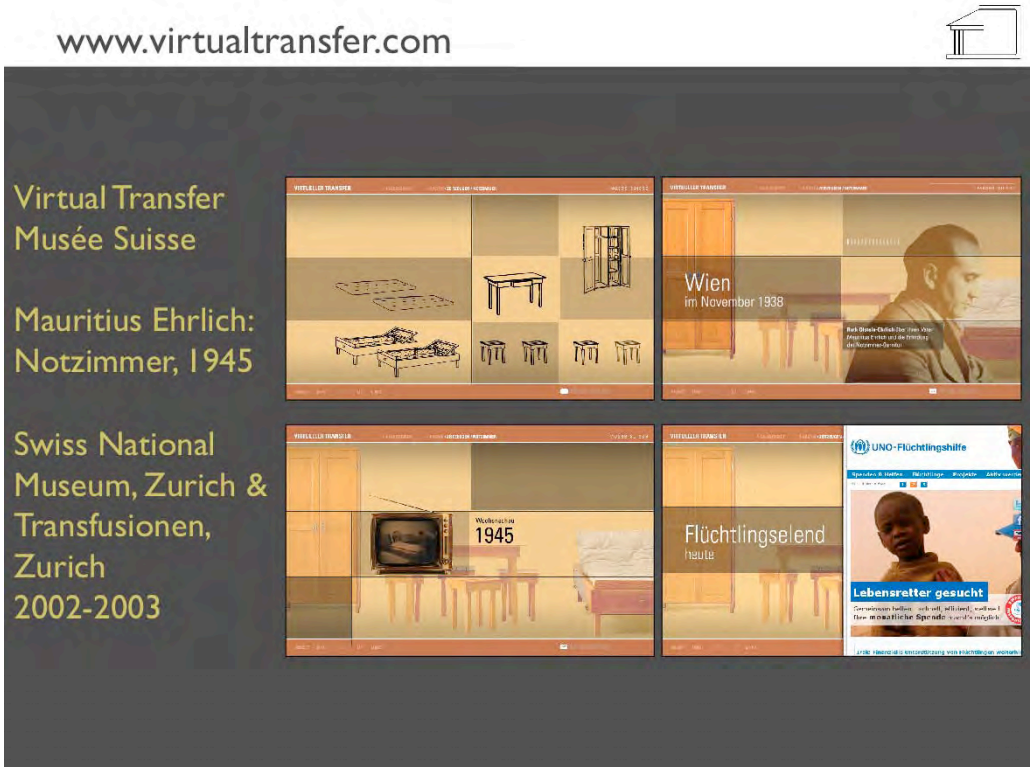



One of the reasons why I've started to produce hypermedia applications has been to learn more about the process of making. But I was not only interested in the production management, more in the theory behind it.

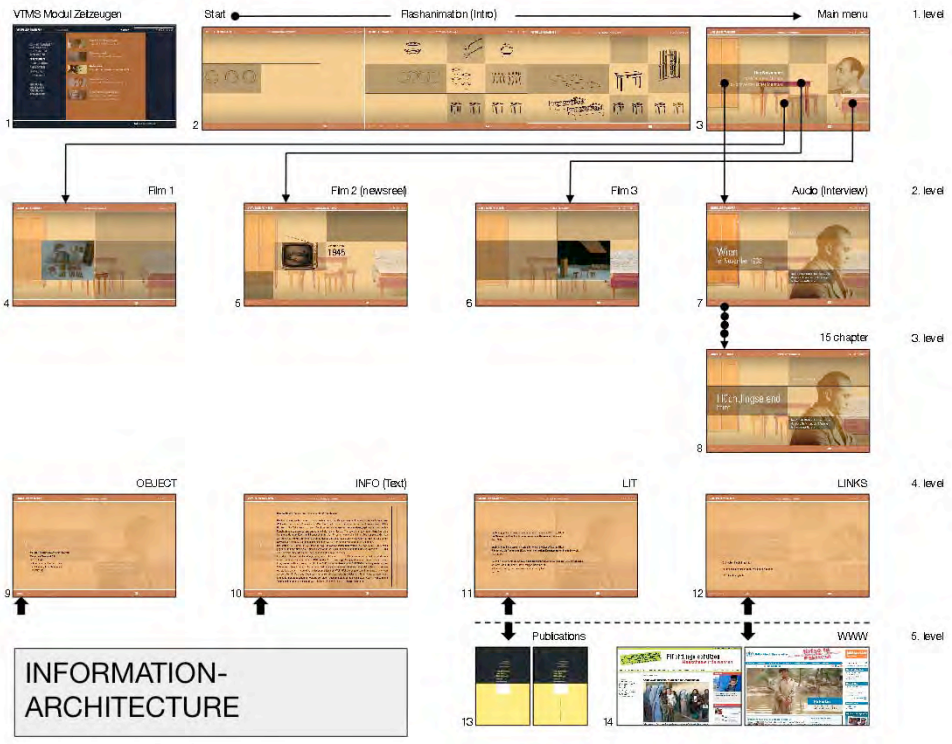
On the left side you will find the different aspects of the analysis give by Garrett, Olsen and Khazaeli.

And on the right side is my own structure for the analysis.

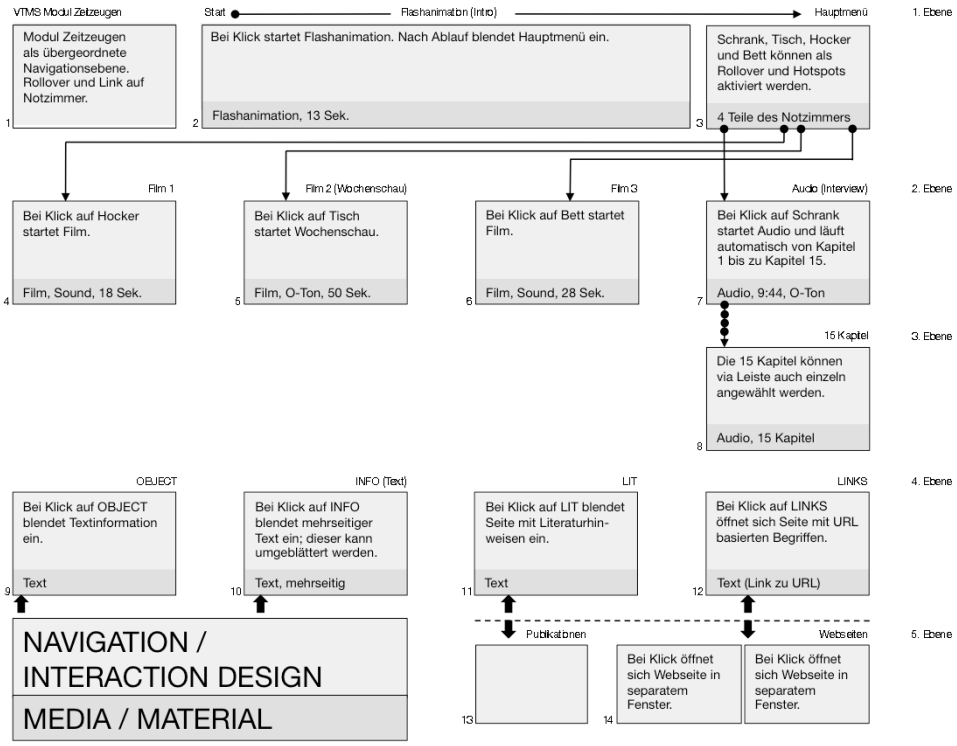
As you can see is the process of analyzing a hypermedia application much more complicated than the creation of it. All in all I have defined over 20 different levels or layers to analyse the complexity of the whole.




Let's have a look of one of the case studies and go back to the Virtual Transfer Musée Suisse Project.

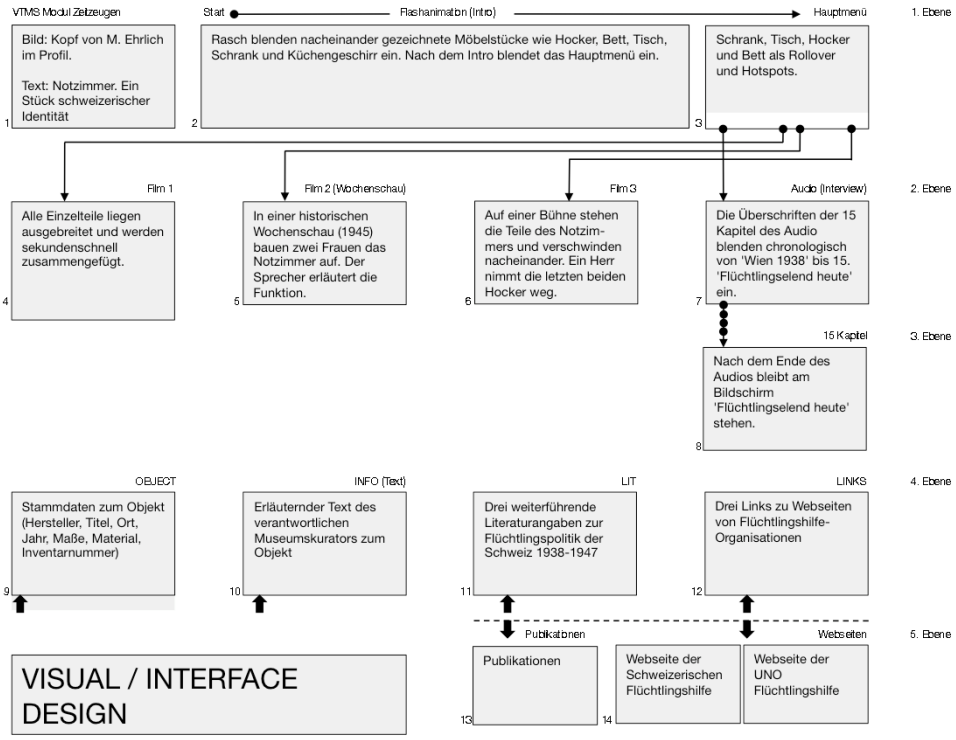



Information Architecture

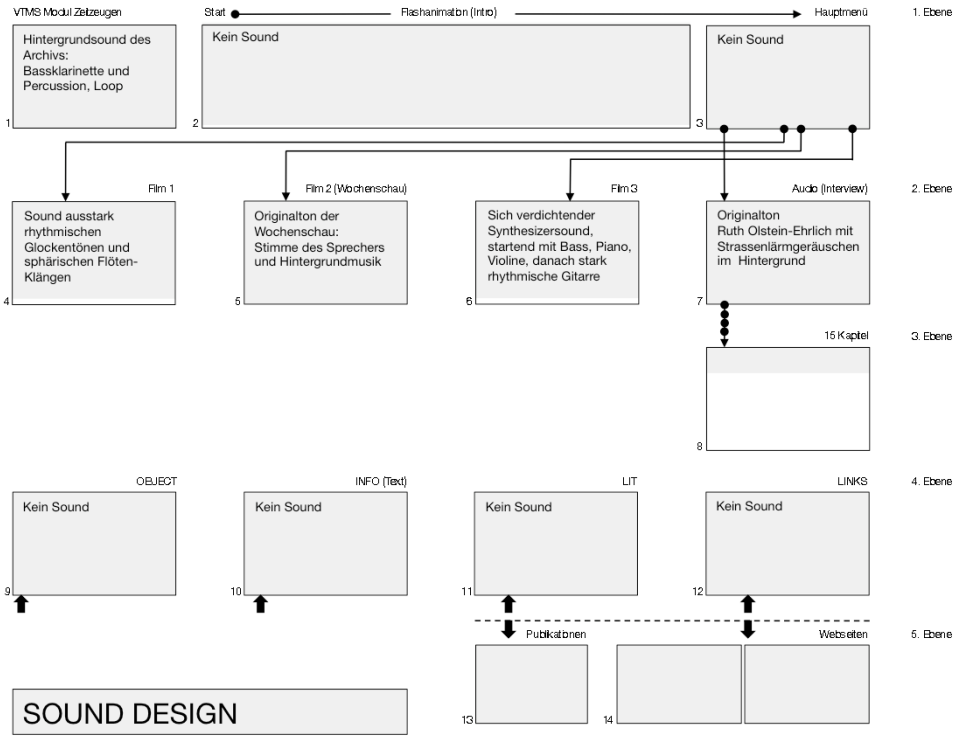



Navigation / Interaction Design

Media / Material




Visual / Interface Design




Sound Design




Mauritius Ehrlich helped refugees by inventing a construction kit for furniture in 1945. The Swiss government supported him to produce 30.000 Notzimmer.




The misery of refugees is timeless. Become a „Mauritius Ehrlich“ in your time and help other people who are in poverty.

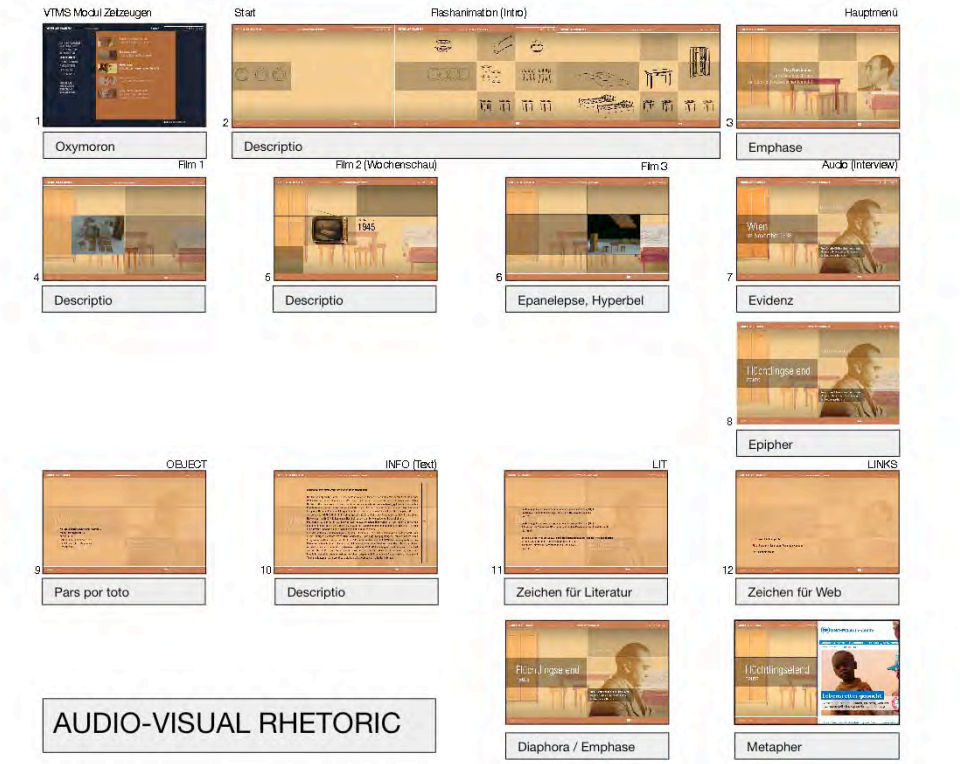
MESSAGE

In the Notzimmer we will find two messages.

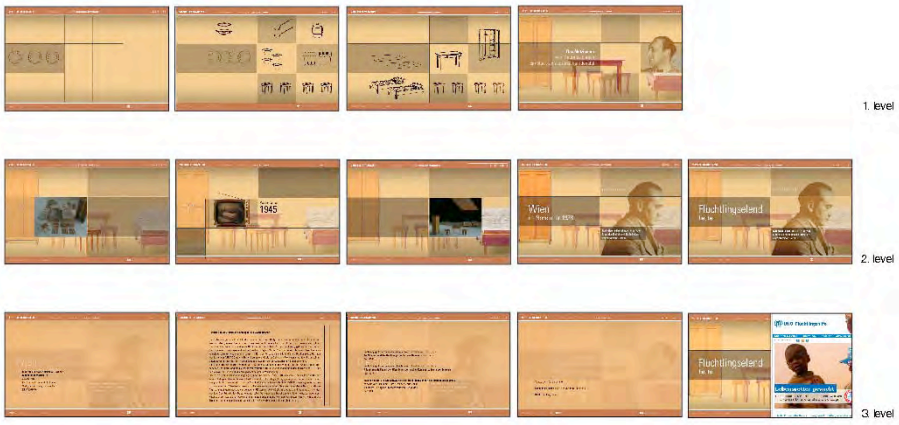
First: Mauritius Ehrlich helped refugees by inventing a construction kit for furniture in 1945. The Swiss government supported him to produce 30.000 Notzimmer.

Second: The misery of refugees is timeless. Become a „Mauritius Ehrlich“ in your time and help other people who are in poverty.

So the artefact changed from a historic document to a powerful statement given by the museum.

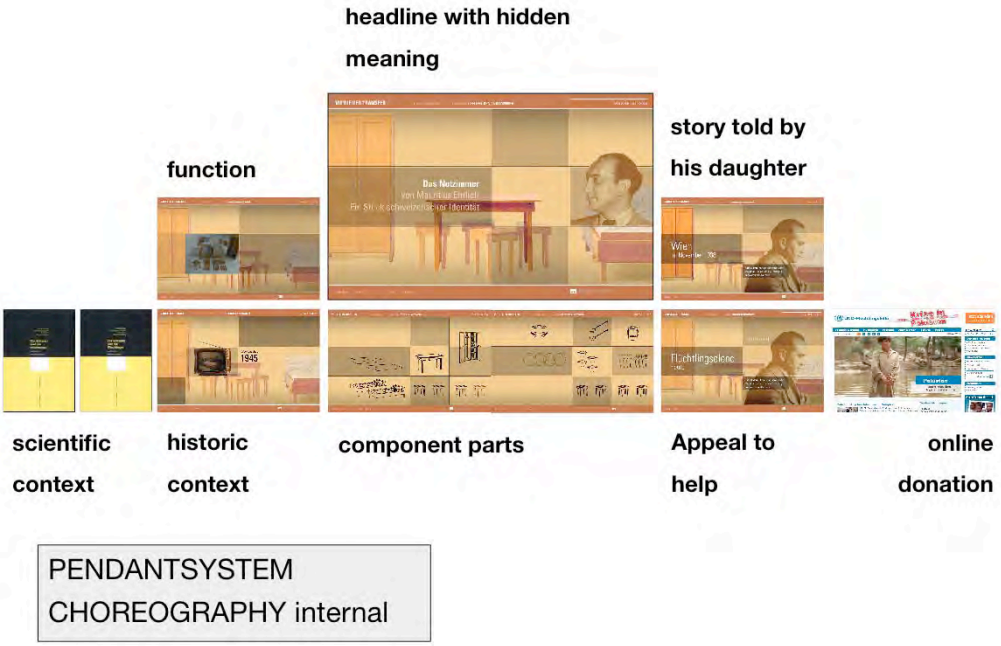



Audio-visual Rhetoric



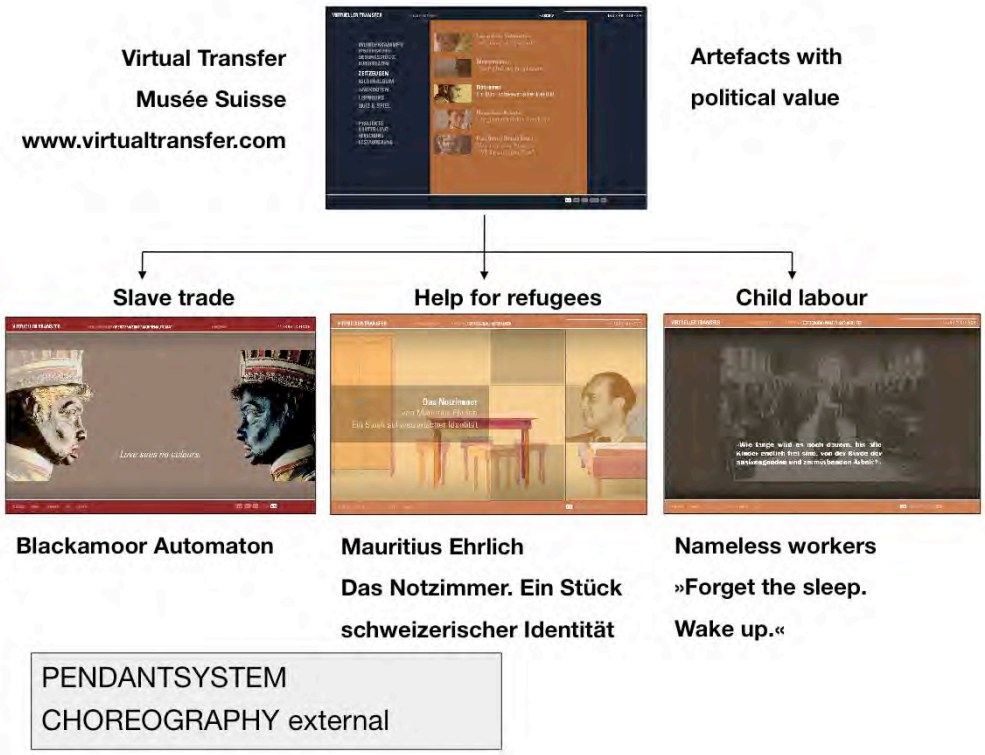

VISUAL GRAMMAR

Visual Grammar




This is the Pendantsystem with the many levels of value and with an internal choreography.

The term "Pendantsystem" – I haven't found an English translation – is a technical name for the hang of paintings in a museum in the 19th century. It characterizes the correlation between the different parts of the hang. The Pendantsystem is a fantastic synonym to explain hypermedia.

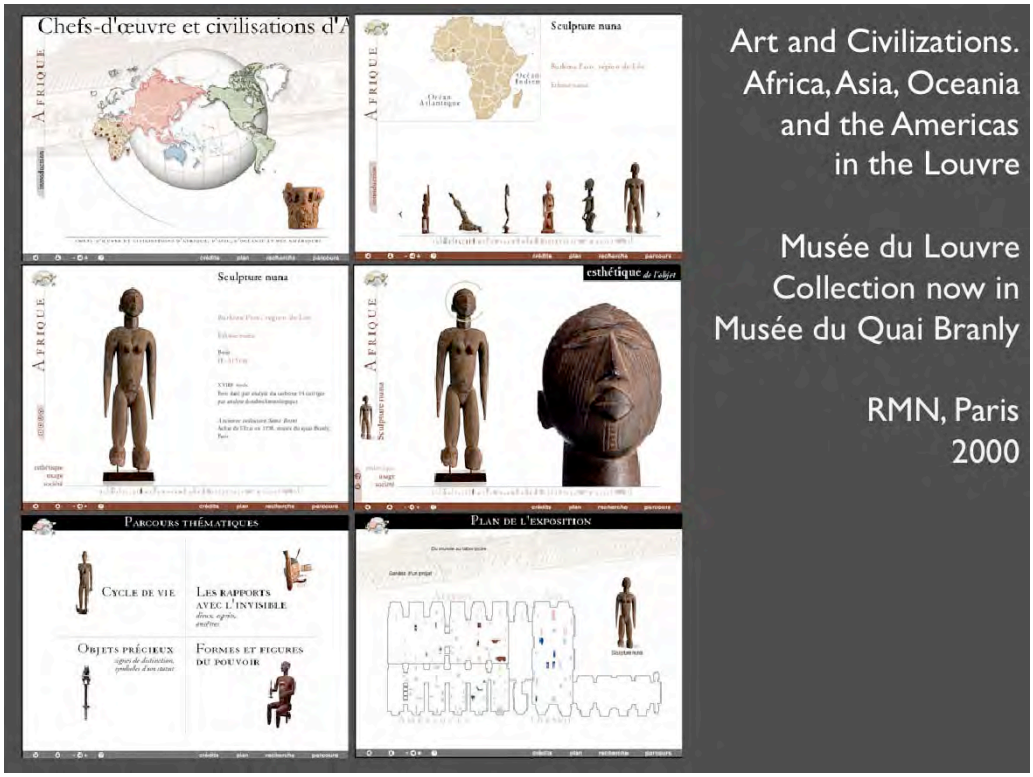



But the Pendantsystem exists also with an external choreography. Hereby our Notzimmer with its message to help refugees will get connected to other artefacts of the collection with political value like the Blackamoor Automaton who is a symbol for slave trade and historical photos with nameless workers standing for child labour.



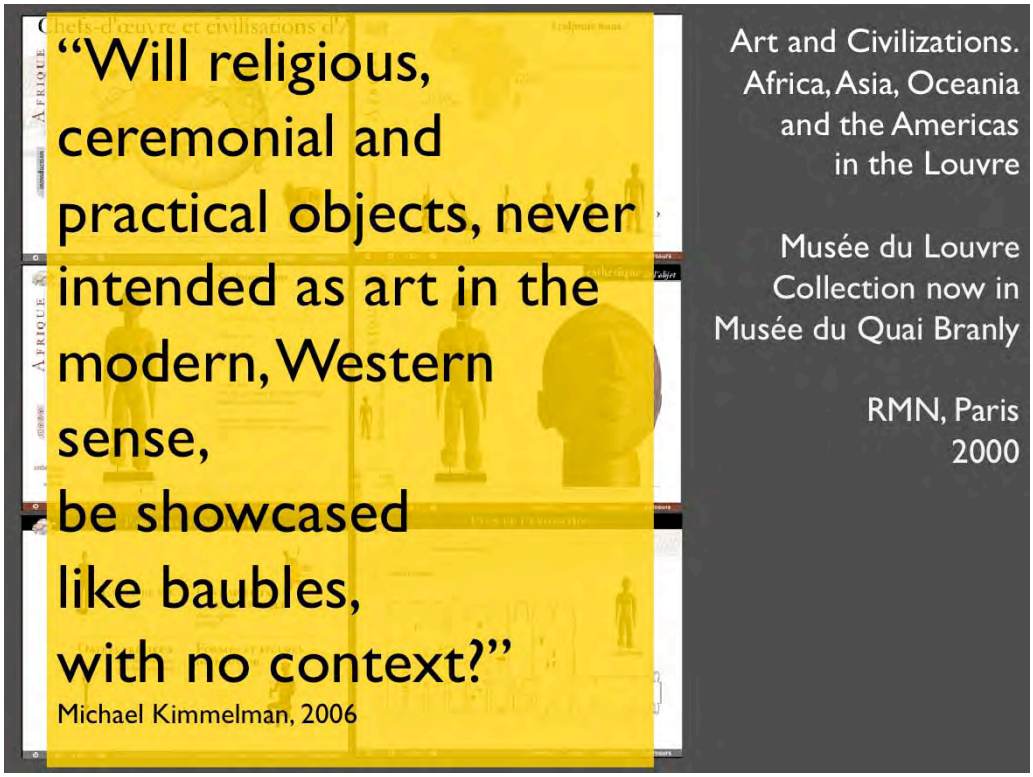
"Design is an art of thought directed to practical action through the persuasiveness of objects and therefore, design involves the vivid expression of competing ideas about life." Richard Buchanan, 1985
"visual-verbal rhetoric" Gui Bonsiepe, 1965
"audio-visual rhetoric" Heike Sperling, 1998

Richard Buchanan anticipated design as a way of rhetoric communication and interaction, when he said in 1985: "Design is an art of thought directed to practical action through the persuasiveness of objects and therefore, design involves the vivid expression of competing ideas about life."
Gui Bonsiepe was the one who drafted in 1965 the elements of "visual-verbal rhetoric" for graphic design.
Gesche Joost and Arne Scheuermann shown in their book "Design als Rhetorik" that rhetoric is used as a model to describe the processes of design. Meanwhile tools of rhetoric help to analyze advertisements, commercials, film, web, as well as interactive multi-media. So the term "audio-visual rhetoric" seems to be predestinate for analyzing the syntax (form) and the semantic (content and meaning) of the full particulars of hypermedia. To explain this, the following example has been described by text and by rhetoric terms.




"Art and Civilizations. Africa, Asia, Oceania and the Americas in the Louvre"
Starting from a global map as the main menu the user can decide which continent s/he would like to explore. What an "analogy" or rather a "metaphor" is the boundless conquest of the world by explorers. A map of the chosen continent and a series of colored (!) artifacts appear at the bottom of the white (!) screen and a second series below with the same artifacts as miniatures. Colored artifacts stand in line in a clean white space. This "accumulatio" as an overview is a "symbol" for the storage of the museum. On the other side we find another "metaphor" for the conquest by colonial power. Clicking on "plan" selects the artifacts and group them by continent. The "plan" of the ground floor shows the formerly position of the artefacts in the Pavilion des Sessions in the Louvre. This part is designed like an inventory. It combines "accumulatio" as "enumeration", and "repetition" with the "accentuation" of the single object.

This CD-ROM is well designed, but what message is hidden behind its visual grammar? In view of the fact that this CD-ROM has a powerful index of all kinds of information, it seems that enthusiastic colonialists and experts in ethnology are the main target groups. It is an oeuvre catalog of the ethnological collection of the Louvre and at the same time a historic document of the formerly installation in the Pavilion des Sessions.




On the other hand, we have a design that shows cultic artifacts in a "white cube." The meaning of those cultic artifacts is often unclear. It seems that every single object is a manifestation of the human ability to materialize the secrets of life. These ethnological artifacts were torn out of their cultural background and environment. Often they were collected under strange circumstances, and transported to France. Some of them (not a few) are victory signs of colonial power. Now the "white cube" of the CD-ROM transforms these cultic artifacts into works of art. During the opening of the Musée du Quai Branly this "esthetics-versus-ethnology" debate was discussed: "Will religious, ceremonial and practical objects, never intended as art in the modern, Western sense, be showcased like baubles, with no context?"

The combination of a "white cube" with a letter case suggests democracy for all artifacts. It seems to be a "pantheon" of museum objects in the wide white space of museum architecture. Purified of the past history of colonialism, the new works of art have lost their meaning as symbols and therefore their semantic power. Now the audience can admire the artifacts as works of art in a politically correct way. The CD-ROM "Art and Civilizations. Africa, Asia, Oceania and the Americas in the Louvre" has used the colonialistic philosophy of the Musée du Quai Branly in advance. And so this CD-ROM appears as an "oxymoron" and transfers with a certain "sarcasm" into a "symbol" of postcolonialistic world outlook.




There is something more to discover than the pure rhetoric elements. This is the complex meaning of the whole hypermedia application or rather the "tone" of the whole. Every application has its own visual, auditory, and interactive tone. As a sequence of different rhythms in the sense of a grammar, it is designed by liquid information architecture and structured by a system of hyperlinks. Let's have an example:

The CD-ROM "Verner Panton" has been produced as four kiosk systems for the exhibition show at the Vitra Design Museum. Following the timeline in the chapter "Biography", the user will get an introduction to the life of Panton, as well as an overview of his oeuvre. The second chapter "Panton Chair" contains detailed information about the technical development and the differences in material, mold, and production of this famous single piece chair. And give us information about it's usage. It is the artefact itself who is the narrator. The third chapter "Interior Design" shows four legendary interiors by Panton. All these projects, like "Visiona", shown in 1970 on a Rhine cruiser, are presented in detail, including plans and photos. Interactive floor plans allow the user to see the interior from different angles of vision. This strategy of interactive voyeurism is like a voyage of discovery back in the 1970s.

Verner Panton
Vitra Design Museum, Basel: Hochschule für Gestaltung und Kunst 2000



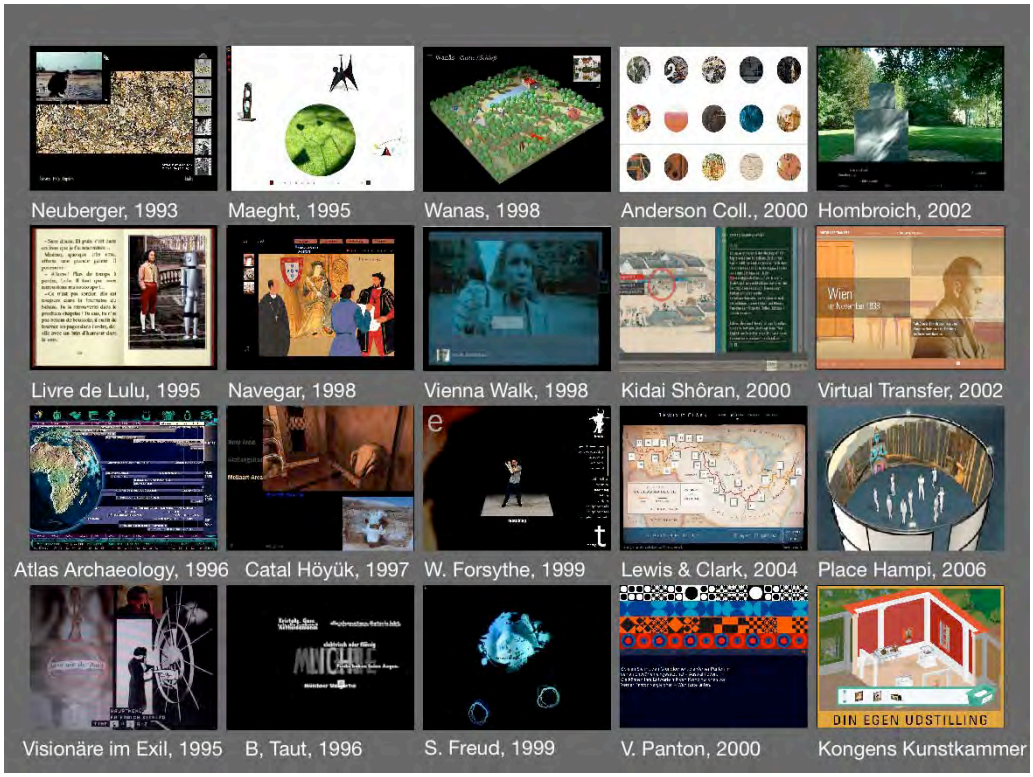

In "Textile Design", the fourth chapter, the user gets an introduction to the foremost creative instruments for the design of textiles: colors and geometric patterns. The screen holds 76 textile patterns, which can be grouped according to thematic considerations like Op Art or 3D. The last chapter "Pattern Game" is a clever supplement to "Textile Design." Four different patterns can be modified by numerous variations and can be compared with the original textile design of Panton himself. So the game teaches the user how to compose ornamental textiles. With its five parts the application about Verner Panton is addressed to all kind of visitor and user groups. The encyclopedist or the expert will find informations. The interested voyeur can make observations. The playful discoverer can play a game. All in all, the user will get a mixture of basic documentation of Panton's philosophy and an incomplete oeuvre catalog of his design in the 1960's and 1970's. Furthermore it is a source book to create new patterns. The CD-ROM about the Danish designer Verner Panton is an excellent example of how dramaturgy, navigation and design gear together in a synergetic way. It will show us that a hypermedia application is more than the sum of its parts. And it is a rare example for the integration of all categories of user.






If the most successful way to transfer information is the "creation of a confidential relation", then the primary objective has become to define strategies by using the visitor's own experiences, preferences, and curiosity. To capture the visitor's "feeling of solidarity" by empathy will be the key to create polyperspective readings and hyperlinked structures. The secret of acceptance is to visualize the visitors hidden needs in a sensitive and inspiring dramaturgy. If this is accepted by the user, viewer or visitor, all hypermedia applications will become successful and typify the merit of education, as Parry and Arbach have shown in their study about online museum learning.

Traditional narratives use a linear structure, pushing a single storyline, while hypermedia applications need a network of hyperlinked mini-stories, targeted at different user-groups and versatile enough to be used in different fields of application. Furthermore, the narratives in hypermedia applications use a type of language combining verbal rhetoric with visual rhetoric to create a "cognitive design."




To unify content, navigation, and design it is necessary for the designer to realize a sort of visual intelligence, but also demands hypermedia competence of the user. Interactivity, the multiple choices of digital one-way streets, encompasses the intention to participate in both: creativity and confrontation. The first signifies a joyful collaboration, the second a clash with the system. Real interactivity thus grows out of the actuality of an action. It is productivity that leads to the creation of reality and the meaning of perception. The best navigation is content in itself and not merely a simplified duplication of the content. Thus information architecture should create different levels of context and animate the user to construct meaning autonomously. Readability is therefore directly dependent on the experiences and personal needs of the user him/herself. Timeless questions need contemporary – in our case – hypermedial answers. Just telling interactive stories is not enough. In a film, each individual sequence is responsible for the rhythm of the story and the dynamics of the actions. The eye of the viewer easily ends up being bored if there is no change of scene and no variety to the shots. To subdue the visual ennui a diversified and eventful dramaturgy is necessary. The real challenge of such a dramaturgy is going to be to find a way to design motivating modules of content with an overall visual and acoustic concept.

## How can we catch the attention of the user?

## How can we decelerate the dynamics of the user attitudes?

## How can we simplify the complexity of the information without losing its value?

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Dramaturgy in a film is the correct application of pacing, of rapid pulse beats, and then pausing for a moment. This means that the story is not only used as a narrative but also as a way of design, and furthermore that the story serves as a means for the recipient to identify with the content. To transfer this strategy into the liquid architecture of hypermedia means to understand the principle of simultaneity in non-simultaneity.

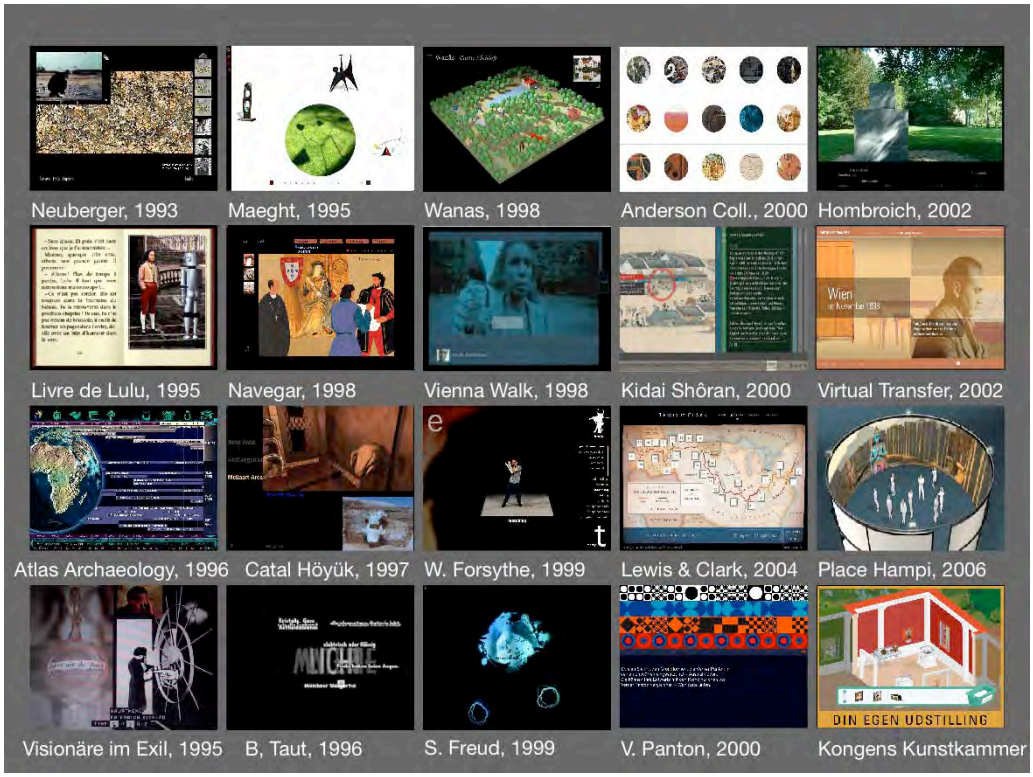
"How can we analyze the structure of such
a hybrid experience, and
how can we better design this kind
of complex space?
How do we integrate media rich environments
with people-rich ones and make them
human, warm and conducive to learning?
How do we organize these experiences
for the user so they can make sense of them
without robbing them of their inherently
rich and spontaneous qualities?"
Larry Friedlaender, 1999

Larry Friedlander has shown us that the problems encountered when we try to communicate information about art are not art-specific but essential for the common transfer of knowledge.

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How do we integrate media rich environments with people-rich ones and make them human, warm and conducive to learning?

How do we organize these experiences for the user so they can make sense of them without robbing them of their inherently rich and spontaneous qualities?"




Consequently it is important to learn how to accelerate and decelerate the flow of information at the right moment, as well as learning how to lead the user with a storyline and equally through a navigational design. It should be possible to create a work that is at the same time complex yet still intuitive.

In my talk some of the provocations for a new grammar of hypermedia has been shown. A typology containing all possible categories has to document what has happened as well as what will follow. Still yet documentation is retrospective. The numerous means of hypermedia design allow an imaginative, innovative, inventive information design, which is meant to be intuitive in its usage. Any new project still needs to define its own rules in accordance with the goals of the said project. In this interplay of content and dramaturgy, navigation and design, lie the challenges for the co-determination of the hypermedia future and for the documentation.



空穴来风 未必无因

„If the wind comes
from an empty cave,
it‘s not without a reason.“

Let’s end with the hidden wise secret behind hypermedia:

"If the wind comes from an empty cave, it‘s not without a reason.“

谢谢大家

Thank you for your attention.

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